

Shay Mordechai

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Summary

Offensive Security Researcher specializing in Android internals, vulnerability discovery, and malware triage. Recognized for uncovering security-critical compiler bypasses (CERT-IL #11265) and authoring comprehensive research on OS execution boundaries. Proficient in ARM64 reverse engineering, AI-assisted fuzzing, and dynamic instrumentation (Frida), with a strong focus on dissecting multi-stage attack chains across Java, C, and Assembly.

Skills

Reverse Engineering | Android Internals | Frida (DBI) | JADX | ARM64 Assembly | IDA Pro | Vulnerability Research | Fuzzing Automation | Threat Modeling | Memory Models (V8/JIT) | Python | C/C++ | Java

Professional Experience

Application Security Intern, Israel Tax Authority | Aug 2024 – Jul 2025

- Performed web application security analysis (OWASP Top 10), tracing execution paths to identify authentication and logic flaws, reducing the attack surface in production environments.
- Orchestrated the deployment of SAST tools (Checkmarx) across 15+ environments, enabling the proactive mitigation of application-layer vulnerabilities.

Combat Intelligence Specialist | IDF (Hesder) | 2016 – 2018

- Specialized in identifying "low-signal" anomalies and tracking adversary methodologies using data-driven analysis under high-pressure conditions.

Projects & Research (Detailed Write-ups at shay-mordechai.github.io)

- Android Malware Triage Research (Self-directed):** Developed a structured 5-layer malware analysis framework covering static triage, business logic camouflage detection, dynamic code loading, evasion patterns, and chained attack flows across Java/Android, JavaScript, C, and Assembly. Completed OWASP UnCrackable L1-L2 challenges using Frida instrumentation.
- Mobile Internals & Execution Boundaries (Publication Series):** Authored a 5-part technical series ("The Paradox of Mutability") analyzing security boundary degradation across OS layers. Analyzed Android execution flows (Java/JNI/Native) and bypassed OS-level anti-debugging mechanisms (ptrace, integrity checks) using Frida.
- Compiler Security & Fuzzing Automation (SBCL):** Architected an AI-assisted Python fuzzing harness to systematically explore variable isolation boundaries during macro expansion. Identified lexical scope bypasses via dynamic binding manipulation (`proclaim`), demonstrating strong threat modeling and automated vulnerability assessment methodologies.
- Agentic AI Threat Research:** Deployed a Cloudflare Worker honeypot to analyze AI browser behavior; successfully identified semantic filter bypasses and exposed Azure ASN fingerprinting of OpenAI agents.
- Security Architecture & Engineering (MyLeads AI):** Engineered a Wasm-based DLP filter for Envoy to redact sensitive egress data and hardened production environments using Rootless Podman and Linux namespaces.

Education

B.Sc. Computer Science | Jerusalem College of Technology | 2021 – 2025 (GPA: 89) - Core Focus: *Core Focus:* Reverse Engineering, Advanced OOP (Java, C++), Malware Analysis, Network Protocols.